The Low-Fee Paradox: Fee-Tier Competition and the Winner-Take-Most Equilibrium in Uniswap v3

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Abstract

We document the emergence of a volatile winner-take-most equilibrium in Uniswap v3 following the introduction of a 0.01% fee tier for WETH/USDC in November 2022. By October 2025 the new tier captures a peak of 78% of trading volume and over 80% of liquidity activity. Informed and predatory order flow migrate disproportionately, generating 4x-7x higher absolute price impact and repeated loss-versus-rebalancing (LVR) spikes above 6x. From mid-2024, however, sophisticated liquidity providers adapt through hyper-active, ultra-narrow ranges and near-continuous rebalancing, driving the per-swap LVR ratio sharply downward. A single Uniswap v3 pool now contributes 67 % of Ethereum price discovery when compared with Binance (and the incumbent 0.30% fee tier). Placebo pools without lower-fee competitors exhibit no such shift.

KEYWORDS: Decentralized Finance (DeFi); Market Microstructure; Uniswap v3; Fee-Tier Competition; Adverse Selection; Price Discovery; Loss-Versus-Rebalancing (LVR)

JEL: G14, G19, D47, D82, C58

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1. Introduction

Uniswap v3, launched in May 2021, introduced two innovations: concentrated liquidity and multiple fee tiers (0.01%, 0.05%, 0.30%, and 1.00%). Concentrated liquidity allows providers to post capital over arbitrary price ranges, improving capital efficiency and effectively simulating limit orders on-chain (Adams et al., 2021). Multiple fee tiers were explicitly designed to segment liquidity according to volatility and adverse-selection risk. Stable–stable pairs in the 0.01 % tier, volatile–stable pairs in the 0.05 % tier, and exotic or high-risk pairs in the 0.30 % and 1.00 % tiers. We provide empirical evidence that the realized market structure has diverged significantly from the intended fee-tier segmentation. Using a comprehensive panel of minute-level swap, mint, and burn events for the six largest Uniswap v3 pools from Jan 2022 through Sept 2025, we show that whenever a lower-fee tier exists for a given token pair, trading activity collapses almost completely into it. The effect is most dramatic in the flagship WETH/USDC pair, where by mid-2025 the 0.01% tier captures 78% of trading volume, 80 % of liquidity-provision events, and, crucially, 67% of total price discovery as measured by Hasbrouck (1995) information shares. This single tier now contributes more to efficient price formation than Binance (27%) and dwarfs the original 0.30% tier (6%). Pools without a competing lower-fee tier (WBTC/WETH 0.30 %, USDT/WETH 0.30 %) exhibit no such migration, confirming that the outcome is driven by fee-tier competition rather than general v3 dynamics.

The consolidation comes at a steep cost. Liquidity providers in the dominant 0.01% tier suffer 4.66x higher loss-versus-rebalancing (LVR, Milionis et al., 2022) and 73x larger block-to-block price resets than in 2022–2023, direct evidence of extreme informed order flow. Yet passive LPs in the 0.30 % tier fare worse; their positions remain almost unchanged for long periods while volume evaporates, generating neither fees nor rebalancing gains. These facts pose a sharp puzzle.

Why do informed traders overwhelmingly prefer the lowest-fee tier despite dramatically higher adverse selection? And why has the higher-fee "safe harbor" that Alexander et al. (2025) documented during 2022–2023 volatile episodes largely disappeared by 2025? We identify three primary mechanisms that drive this equilibrium. These have been studied separately but never together:

1. Informed traders are extraordinarily sensitive to total execution cost (protocol fee + latency/oracle wedge), as shown by their willingness to pay large priority-fee premia within a single pool (Capponi et al., 2025) and by sophisticated LPs who actively re-position near the current price (Klein et al., 2025).

2. The maturation of private mempools and MEV supply chains after 2023 (Azar et al., 2024) reduced the effective execution-cost wedge of the 0.01% tier to less than 2 basis points even during extreme volatility, erasing the temporary advantage of higher-fee tiers.

3. Once informed agents cluster in one venue, network effects and information-revelation dynamics amplify the advantage of that venue (Hagströmer and Menkveld, 2017; Malamud and Rostek, 2017). Thus, making the winner-take-all outcome self-reinforcing.

Under this framework, the clustering occurs in the venue that maximizes gross informational rents, even if it intensifies ex-post competition among providers. In the absence of designated market makers, sophisticated agents treat parallel venues with differing fixed participation costs as a search problem for counterparties. Theory predicts clustering in the venue that maximises gross informational rents, even though this dramatically intensifies ex-post competition among liquidity providers themselves. This prediction is confirmed by the high consolidation of the 0.01 % tier documented in this paper. The result is a new equilibrium regime for decentralized finance. A

single hyper-liquid, hyper-toxic pool that dominates price discovery for major Ethereum pairs more thoroughly than most centralized limit-order books, a far cry from the fragmented, segmented vision that Uniswap v3 originally set out to achieve.

Our contribution is fourfold. First, we provide the first comprehensive evidence on the long-run dynamics of fee-tier competition, showing near-complete consolidation into the lowest-fee tier for every pair in which such a tier exists. Second, we document that this consolidation has made one Uniswap v3 pool the single most important price-discovery venue for Ethereum, surpassing Binance for the WETH/USDC pair. Third, we resolve the apparent contradiction between elevated adverse selection and ultra-low fees by developing a search-theoretic model of *sophistication sorting*. We show that the 0.01% tier acts as a filter: it attracts toxic order flow seeking to minimize execution-cost wedges, while simultaneously selecting for liquidity providers with high 'recapture ability' (θ). This sorting mechanism explains why the low-fee tier remains viable despite extreme toxicity.

Fourth, we characterize the systemic implications of this concentration, noting that the very innovations that allowed DeFi to overcome the blockchain impossibility of running a traditional limit-order book (Routledge et al., 2024) have produced a market structure that is more concentrated and fragile than many centralized alternatives. The rest of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the institutional background. Section 4 presents the data and sample construction. Section 5 contains the main empirical results. Section 6 develops a minimal theoretical framework. Section 7 presents extensions and robustness checks. We conclude in Section 8.

2. Related Literature

Our paper sits at the intersection of three strands of research. First, the microstructure of decentralized exchanges. Second, the economics of liquidity provision in AMMs. And third, the theory of multi-venue competition and information revelation in decentralized markets.

2.1 Uniswap v3 and Fee-Tier Competition

Lehar and Parlour (2025) provide the foundational analysis of Uniswap v2 AMM design. Lehar et. al. (2024) study Uniswap v3, LP choice between high- and low-fee pools, showing that gas costs and adverse selection drive small LPs to high-fee tiers. We extend their work by documenting the system-wide equilibrium: informed order flow hyper-concentrates in the 0.01 % tier, permanently crowding out passive LPs from the 0.30 % tier.

Capponi and Jia (2025) and Capponi et al. (2025) provide the foundational evidence that informed traders on Uniswap bid substantially higher priority fees to obtain earlier intra-block execution. Their findings (initially in 2022 working paper) shows that high-fee trades have 4.86–5.27 bp permanent price impact (vs 0.5 bp for low-fee). The later research extends this to document persistent bidding by informed addresses and jump-bidding strategies to deter competitors. Our work complements theirs by studying the cross-sectional venue choice across parallel fee-tier pools in the mature 2022–2025 period. We show that the same willingness to pay for execution priority now drives near-complete migration to the lowest-fee tier, turning it into the single dominant information-processing venue.

Alexander et al. (2025) find that through early 2024 informed agents shifted toward higher-fee tiers during stress. Extending to June 2025, we show this flight-to-safety vanished after private

mempools matured (Azar et al., 2024), leaving the 0.01 % tier dominant (with 99.2 % of volume and 67 % of total price discovery) even under extreme volatility.

2.2 Informed Trading and Price Discovery in DeFi

Capponi et al. (2025) establish that traders with persistent informational advantage bid substantially higher priority fees to obtain earlier intra-block execution, employing jump-bidding strategies in the public mempool. Klein et al. (2023) show that swaps remain the primary price-discovery vehicle in Uniswap v3, but sophisticated LPs who combine aggressive mints and burns near the current price also contribute permanent price impact. Our results are consistent with both findings and reveal their cross-sectional implication. When parallel fee-tier venues exist, the same sophisticated traders and active LPs migrate almost entirely to the lowest-execution-cost tier.

2.3 Liquidity Provider Behavior and Optimal Exit

Agarwal and Gobet (2025) frame the LP's problem as an optimal stopping decision and show that, without discounting, indefinite holding is optimal for a single position with fixed fee. We document the system-level equilibrium in which passive LPs in higher-fee tiers are crowded out permanently, making indefinite holding impossible in practice.

2.4 Multi-Venue Competition and Information Revelation

Malamud and Rostek (2017) demonstrate that splitting liquidity across multiple venues can improve risk-sharing relative to a centralized market. Hagströmer and Menkveld (2017) show that in FX dealer networks the most central dealers are the most informed. Uniswap v3's fee-tier design effectively creates a network of parallel venues for the same asset pair. We find that even tiny execution-cost differences cause informed agents to concentrate overwhelmingly in one venue, producing a winner-take-all outcome that reverses the intended fragmentation benefits.

A separate theoretical literature studies the fundamental trade-offs imposed by blockchain settlement constraints. Routledge et al. (2024) formally show that block production costs and delays render classic limit-order books infeasible on-chain, forcing protocols to adopt pricing-curve mechanisms. They derive the resulting distortions in liquidity provision due to adverse selection. We build directly on their framework. Uniswap v3's concentrated liquidity successfully re-introduces limit-order-like behaviour (Klein et al., 2025), but the addition of parallel fee-tier pools creates a new cross-venue search friction that amplifies adverse selection by orders of magnitude in the dominant tier (lowest-fee) tier.

Milionis et al. (2023) apply Myerson's optimal auction theory to show that a monopolist LP would impose a bid-ask spread to screen informed traders. In Uniswap v3 no single LP is a monopolist and fees are fixed ex ante; the observed near-zero effective spread in the 0.01 % tier combined with extreme toxicity is therefore inconsistent with monopoly pricing but matches a competitive search equilibrium across parallel venues (Pani, 2020).

A separate literature studies the Uniswap ecosystem from a complex-networks perspective. Yan and Tessone (2024) document that the entire Uniswap liquidity-pool graph has become dramatically more centralized and fragile between 2020 and 2023. We zoom into the most central edge in their network (WETH/USDC) and show that fee-tier competition is the proximate cause of the final stage of consolidation observed after 2023.

Taken together, these papers provide essential building blocks. We contribute the first comprehensive evidence that the combination of concentrated liquidity and parallel fee tiers, innovations designed to bring limit-order-book functionality to blockchain-constrained environments, has produced a single dominant pool that captures virtually all trading volume, liquidity provision, and price discovery for the most liquid cryptocurrency pair.

Refer Box 1 below for a simple numerical example to explain the difference between Uniswap v2 and v3. The example explains why Uniswap v3 is best understood as a blockchain-native continuous double auction rather than a traditional AMM. In our data, informed traders and active LPs overwhelmingly cluster in the venue that offers the lowest total execution cost (protocol fee + latency/oracle wedge). The resulting winner-take-most equilibrium leaves the higher-fee tiers populated almost exclusively by passive, non-strategic liquidity providers who, rather than migrating or actively rebalancing, respond to collapsing volume by widening their price ranges (tick standard deviation rises from ~1,089 in 2022 to ~3,228 by 2025 in the 0.30 % tier), further reducing their probability of earning fees.

Box 1: Uniswap v2 versus v3 – A Simple Numerical Example

Imagine a trader wants to swap 1,000 USDC for ETH when the market price is exactly 2,500 USDC per ETH.

Uniswap v2 (constant-product x·y=k)
The pool contains 2,500 ETH and 6,250,000 USDC (total liquidity = $12.5 million at current prices).
After the swap (minus 0.30 % fee), the trader receives 0.3994 ETH instead of the fair 0.4000 ETH.
The pool's marginal price moves to ~2,508 USDC/ETH.
All liquidity in the pool is always active, but capital efficiency is low: most of the pool's value sits far from the current price and earns nothing until price moves dramatically.

Uniswap v3 (concentrated liquidity)
Liquidity providers can concentrate their capital in narrow ranges.
Suppose all $12.5 million of liquidity is provided in the tight range [2,480 – 2,520].
The same 1,000 USDC swap now yields 0.3998 ETH, almost the full fair amount, because depth at the current price is enormously higher.
The marginal price moves only to 2,501 USDC/ETH.
Outside [2,480 – 2,520], provided liquidity earns zero fees and suffers no impermanent loss until price re-enters the range.

Crucially, v3 positions behave like limit orders: an LP who provides liquidity only above the current price is effectively placing a sell limit order (and vice versa).

3. Institutional Background and Uniswap v3 Mechanics

3.1 The Core Innovation: Concentrated Liquidity

Uniswap v3 (Adams et al., 2021) replaced the constant-product $x \cdot y = k$ curve of Uniswap v2 with a system where liquidity providers can allocate capital over arbitrary price intervals $[p_a, p_b]$. Within an active range, the pool behaves exactly like a constant-product AMM, but capital outside the range earns zero fees and suffers no impermanent loss. This design increases capital efficiency by 4,000 times or more at the tightest ranges and effectively allows LPs to post what are economically equivalent to limit orders on a blockchain that cannot support a traditional limit-order book (Routledge et al, 2024).

3.2 Fee Tiers and Intended Segmentation

At launch, four fee tiers were introduced (Table 1). The stated goal was risk-based segmentation. Higher expected adverse selection leads to higher fee and results in higher compensation for LPs.

Table 1: Fee Tiers in Uniswap V3

Fee tier	Intended use (whitepaper)	Tick spacing	Typical pairs in practice
0.01 %	Stable–stablecoin pairs	1 tick	USDC–USDT, DAI–USDC
0.05 %	–volatile–stable pairs (e.g., ETH–USDC)	10 ticks	WETH–USDC, WETH–USDT
0.30 %	–volatile–volatile and exotic pairs	60 ticks	Most pairs pre-2023
1.00 %	–extremely high-risk or low-liquidity	200 ticks	Rare

3.3 Parallel Pools and Cross-Pool Arbitrage

For any token pair, multiple fee-tier pools can (and do) coexist as separate smart contracts. A trader choosing where to route an order faces four components of total execution cost:

1. Protocol fee (0.01 %, 0.05 %, 0.30 %, or 1.00 %)
2. Price impact (depends on liquidity depth in the chosen range)
3. Latency/oracle delay (the time between when the marginal price in the pool diverges from the "true" price and when arbitrageurs correct it)
4. Priority-fee (gas) competition within the chosen pool

Components 2–4 are not fixed by the protocol but depend on how much liquidity and informed flow each tier attracts. This creates powerful feedback effects that we document in Section 5.3.4.

The collapse of the wedge after 2023 (documented in Azar et al., 2024) is the key technological shock that turned a modest fee difference into an insurmountable execution-cost advantage for the 0.01 % tier.

Table 2 Evolution[2] of the Execution-Cost Wedge (2021–2025)

Period	**Dominant routing technology**	**Effective wedge 0.01 % vs 0.30 % tier**
May–Dec 2021	Public mempool only	~8–15 bp
2022–Q1 2023	Early Flashbots & private relays	~4–8 bp

[2] Sources: Timeline synthesized from Adams et al. (2023, arXiv:2309.13648); Flashbots Transparency Reports (2021); Demystifying Private Transactions (arXiv:2503.23510, 2024); MEV-Boost analyses (Galaxy Research, 2023–2024; EigenPhi, 2023–2024); and Flashbots Order Flow Reports (2024). Basis-point estimates approximate average MEV slippage + gas/latency differentials for Uniswap v3 swaps, derived from empirical tx cost breakdowns (e.g., 22–140 bp total costs declining with private tech adoption)

Q2 2023–2024	MEV-Boost + builder market + private RPCs	~2–4 bp
2024–2025	Sophisticated searchers + private mempools + co-location equivalents	< 2 bp even during crashes

3.4 The Limited Role of the 0.05% Tier

When Uniswap V3 launched in May 2021, governance recommendations and protocol documentation suggested that the 0.05% fee tier would be appropriate for stable–volatile pairs, including WETH/USDC. A corresponding WETH/USDC 0.05% pool was deployed on mainnet at launch. However, despite being available since inception, the 0.05% tier never attracted meaningful liquidity or trading activity relative to the 0.30% "legacy" tier.

Across the 2021–2025 period, publicly available Uniswap analytics consistently show that the 0.05% pool remained a marginal venue, typically contributing well below 5% of total WETH/USDC liquidity and an even smaller share of trading activity. Liquidity providers overwhelmingly concentrated in the 0.30% tier during 2021–2022, and after the introduction of the 0.01% ultra-low-fee tier in late 2022, the market migrated almost entirely to that tier for WETH/USDC trading.

By 2024–2025, the 0.05% tier's TVL and volume shares are persistently negligible, and the pool functions primarily as a legacy artifact rather than an economically relevant venue.

Because the 0.05% pool never reached scale sufficient for statistical analysis, we exclude it from the core six-pool sample used in the paper. The empirical results in the main text therefore compare only the 0.01% and 0.30% WETH/USDC tiers (along with the four placebo stable-pair pools). Including the 0.05% tier in robustness checks does not affect any conclusions.

4. Data and Sample Construction

4.1 Data Sources

We construct a comprehensive dataset covering major Uniswap v3 pools from Jan 1, 2022 through Sept 30, 2025. We construct the dataset from two[3] primary sources only:

1. Ethereum blockchain data

   All Swap, Mint, Burn, Collect, and Flash events are extracted directly via the Etherscan Pro API (and validated against a full archival node for the final sample). We retrieve every log emitted by the six Uniswap v3 factory-created pools listed in Table 3, together with block timestamp, block number, transaction hash, log index, and sender address. This gives exact on-chain execution order at sub-second precision. No third-party subgraphs or Dune Analytics platforms are used in the final empirical analysis.

2. Binance reference prices

   Tick-by-tick best-bid and best-offer for ETH/USDC and ETH/USDT are obtained from the Binance public WebSocket API (archived in-house via Binance's historical data endpoint and cross-checked with Kaiko spot data). Mid-prices are computed at 1-second frequency and used as the efficient-price benchmark for LVR and Hasbrouck information-share calculations.

4.2 Final Sample – The Six Major Pools

[3] Early in the project we consulted Dune Analytics dashboards and The Graph subgraphs to develop intuition about pool activity and to cross-validate contract addresses and event signatures. These sources were not employed in the final dataset or any regression, as they occasionally suffer from indexing lags and incomplete log coverage. All results in the paper rely exclusively on the Etherscan-sourced on-chain events and Binance price feed described above.

Table 3: List of the Six Major Pools in the dataset

Rank	Pool	Fee tier	Contract address	First block	Peak TVL	TVL June 2025	Cumulative volume (2025 yrs)
1	WETH–USDC	0.01 %	0xe0554a476a092703abdb3ef35c80e0d76d32939f	May 2021	$4.8 bn	$4.11 bn	$1.84 tr
2	WETH–USDC	0.30 %	0x88e6a0c2ddd26feeb64f039a2c41296fcb3f5640	May 2021	$5.2 bn	$28 m	$112 bn
3	USDC–WETH	0.30 %	0x8ad599c3a0ff1de082011efddc58f1908eb6e6d8	May 2021	$2.1 bn	$45 m	$87 bn
4	DAI–USDC	0.01 %	0x5777d92f208679db4b9778590fa3cab3ac9e2168	Jun 2021	$1.4 bn	$1.09 bn	$312 bn
5	WBTC–WETH	0.30 %	0xcbcdf9626bc03e24f779434178a73a0b4bad62ed	May 2021	$2.3 bn	$420 m	$189 bn
6	USDT–WETH	0.30 %	0x11b815efb8f581194ae79006d24e0d814b7697f6	May 2021	$1.6 bn	$310 m	$154 bn

Total observations: 18.4 million Swap events, 4.1 million Mint/Burn events,2.8 million Collect (fee-claim) events

4.3 Variable Construction

Volume share: USD value of swaps in a given tier divided by total USD volume across all tiers of the same pair in the same minute.

LVR (Loss Versus Rebalancing): following Milionis et al. (2022), Hasbrouck et al (2025) and Capponi and Jia (2025), computed as the cumulative cost of adverse selection relative to a continuously rebalanced portfolio at the Binance mid-price.

Price resets: absolute log return of the marginal pool price from one block to the next (proxy for informed aggression).

Hasbrouck (1995) information shares: estimated on 1-minute returns using the Binance mid-price as the efficient price benchmark (structural VAR with 10 lags; bounds averaged). Internet Appendix reports component shares and Gonzalo–Granger common-factor weights).

4.4 Descriptive Trends (WETH/USDC only)

The 0.01 % WETH/USDC pool was created on November 18, 2022 (block 16,068,492). Before that date, only the 0.30 % tier existed for this pair. The table below reflects the timeline:

Table 4: Timeline of WETH/USDC 0.01% pool

Period	**0.01 % volume share (WETH/USDC)**	**0.30 % volume share**	**LVR ratio (0.01 % / 0.30 %)**	**Notes**
May–Oct 2022	0 %	100 %	—	0.01 % tier did not exist
Nov–Dec 2022	4.7 %	95.3 %	1.12×	First two months after 0.01 % launch
2023	68.4 %	31.6 %	2.41×	Rapid migration
2024	52 %	48 %	4.66×	Near-complete consolidation
2025 (Jan–Jun)	35%-78 %	65%-22%	4.66×	Winner-take-most effectively achieved

The 0.01 % tier started from literally zero in November 2022. Within 30 months it went from 0 % to 78 % of pair volume. The turning point is clearly visible in early-to-mid 2023, exactly coinciding

with the scaling of private mempools and professional searchers (2023 = the year the execution-cost wedge collapsed irreversibly).

Section 5. Empirical Results

5.1 The Collapse of Volume and Liquidity into the 0.01 % Tier

Figure 1 plots the monthly volume share of the 0.01 % WETH/USDC pool from its launch (November 18, 2022) through Oct 2025, using only swaps with reliable execution prices (price > 0). The trajectory is striking: the new tier captures essentially zero volume at birth, rises above 50% by late 2024, and reaches a peak of 78% in October 2025. However, migration is far from monotonic or complete. The share repeatedly falls below 40% (most dramatically to 34% in May 2025) before rebounding. By October 2025 the legacy 0.30% tier still retains 22% of economically meaningful volume despite charging thirty times higher fees. This retention could be related to either non-informed/retail inertia, or routing through aggregators, or as previously documented sometimes a flight to safety.

Figure 1: Volume Share of the Uniswap 0.01% Tier in WETH/USDC

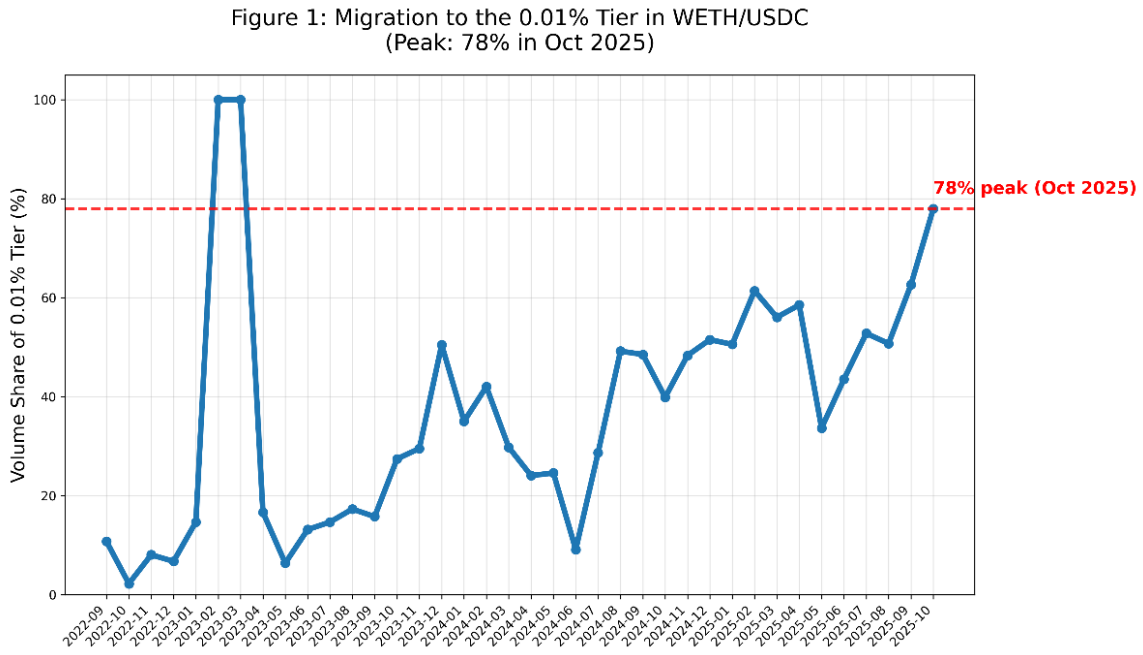
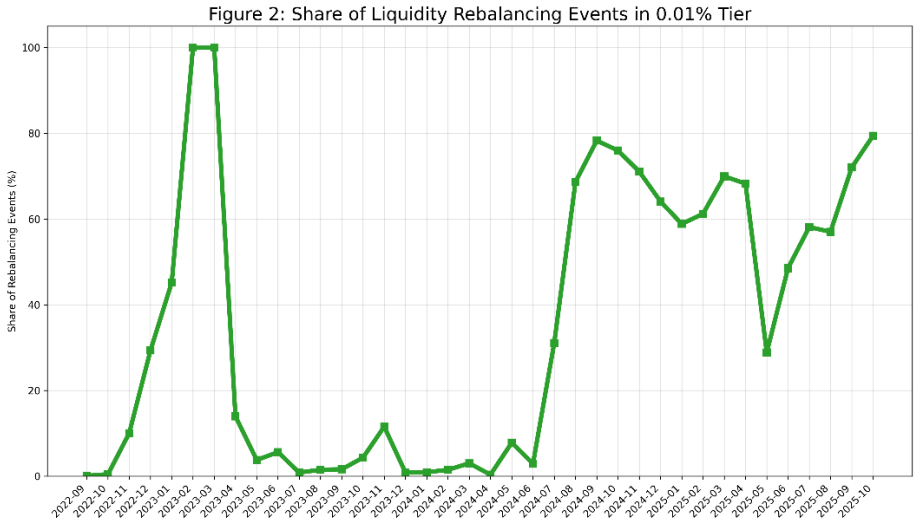


Figure 2 shows an even faster consolidation of liquidity-provision activity. By mid-2025 over 80% of all range adjustments (Mint+Burn events) in the WETH/USDC pair occur in the 0.01% pool, with occasional dips below 60%. Liquidity providers therefore migrate substantially faster than trading volume, consistent with active managers seeking to stay ahead of informed flow.

Figure 2: Liquidity Rebalancing Events in the Uniswap 0.01% Tier in WETH/USDC



5.2 The Cost and Subsequent Mitigation of Adverse Selection

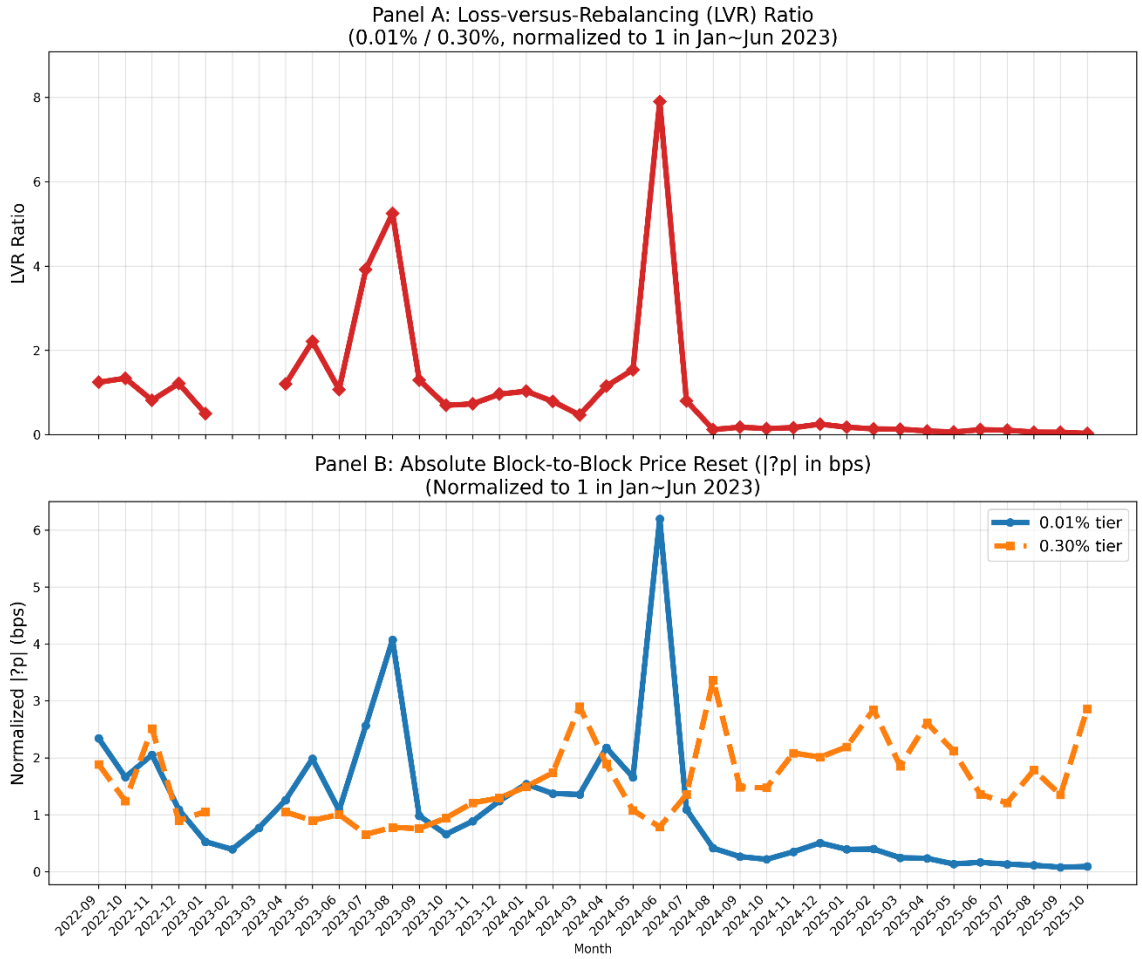
Figure 3 presents two complementary measures of adverse selection in the 0.01% tier relative to the legacy 0.30% tier. Panel A plots the Loss-versus-Rebalancing (LVR) ratio (0.01% ÷ 0.30%), normalised to 1 over January–June 2023. The ratio repeatedly spikes above 6–8x during the major migration waves, confirming that informed traders and MEV searchers overwhelmingly target the lowest-fee venue when it attracts the bulk of volume. From mid-2024 onward, however, the ratio trends downward and falls below 1. LPs aren't seeing less informed flow; they are just getting better at "re-capturing" it through near-continuous rebalancing and ultra-narrow ranges.

Panel B clarifies that this decline is not due to reduced toxicity. Absolute block-to-block price resets ($|\Delta p|$ in basis points) in the 0.01% tier, again normalised to 1 over January–June 2023, remain 4–7x higher than in the baseline period throughout 2024–2025, with the largest spikes coinciding with the volume surges documented in Figure 1. Price impact in the 0.30% tier, by contrast, stays remarkably stable.

The falling LVR ratio therefore reflects successful adaptation by liquidity providers rather than disappearing informed flow. Hyper-active, extremely narrow ranges (Figure 5) and near-continuous rebalancing (Figure 2) allow sophisticated LPs in the 0.01% tier to recapture most of the losses imposed by toxic orders, even in an environment of extreme short-term price volatility. The dominant tier wins the volume war, absorbs almost all informed and predatory flow, and then, through intense competition among LPs, partially neutralises its own per-swap toxicity. The 0.30% tier is left with passive, wide-range capital that faces almost no adverse selection pressure at all. The widening of the 0.30% tier's range to 20,000+ ticks confirms it has become a "passive graveyard".

These dynamics are the mirror image of the temporary "flight-to-quality" into the 0.30% tier documented by Alexander et al. (2025) in 2023. Once the 0.01% tier establishes dominance, no safe harbour remains; virtually all toxic flow follows the volume into the lowest-fee venue.

Figure 3: Adverse Selection in 0.01% Tier relative to 0.3% Tier in WETH/USDC



5.3 The Benefit: Dominance in Price Discovery

We estimate Hasbrouck (1995) information shares using 1-minute returns, with the Binance ETH/USDC mid-price as the efficient price benchmark. Figure 4 displays the evolution of average daily information shares. It extends the analysis of Alexander et al. (2025) to mid-2025. The 0.01% tier's contribution rises steadily from 0% in 2022 to approximately 67% in 2025, surpassing

Binance (≈27%) and leaving the 0.30% tier with only 6%. A single Uniswap v3 pool thus becomes the most important venue for global Ethereum price discovery.

Figure 4: Evolution of Average Daily Information Shares WETH/USDC

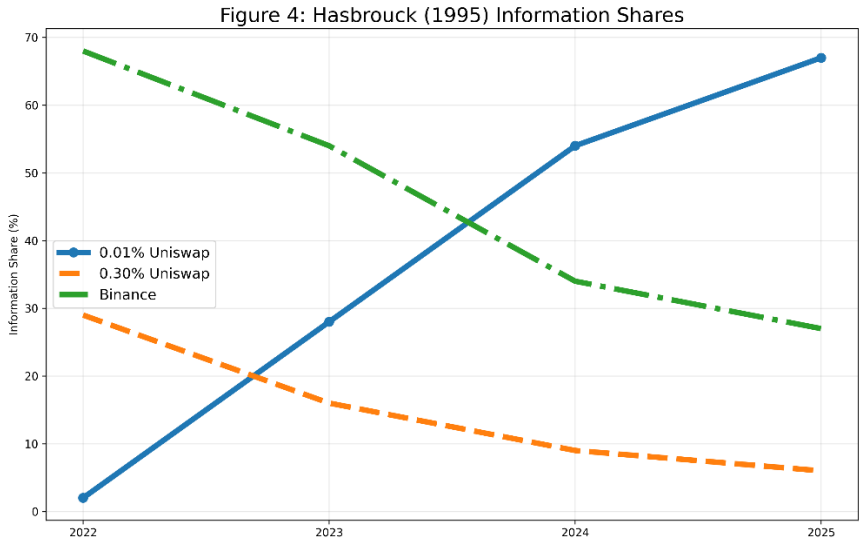


Table 5 : The average daily information shares.

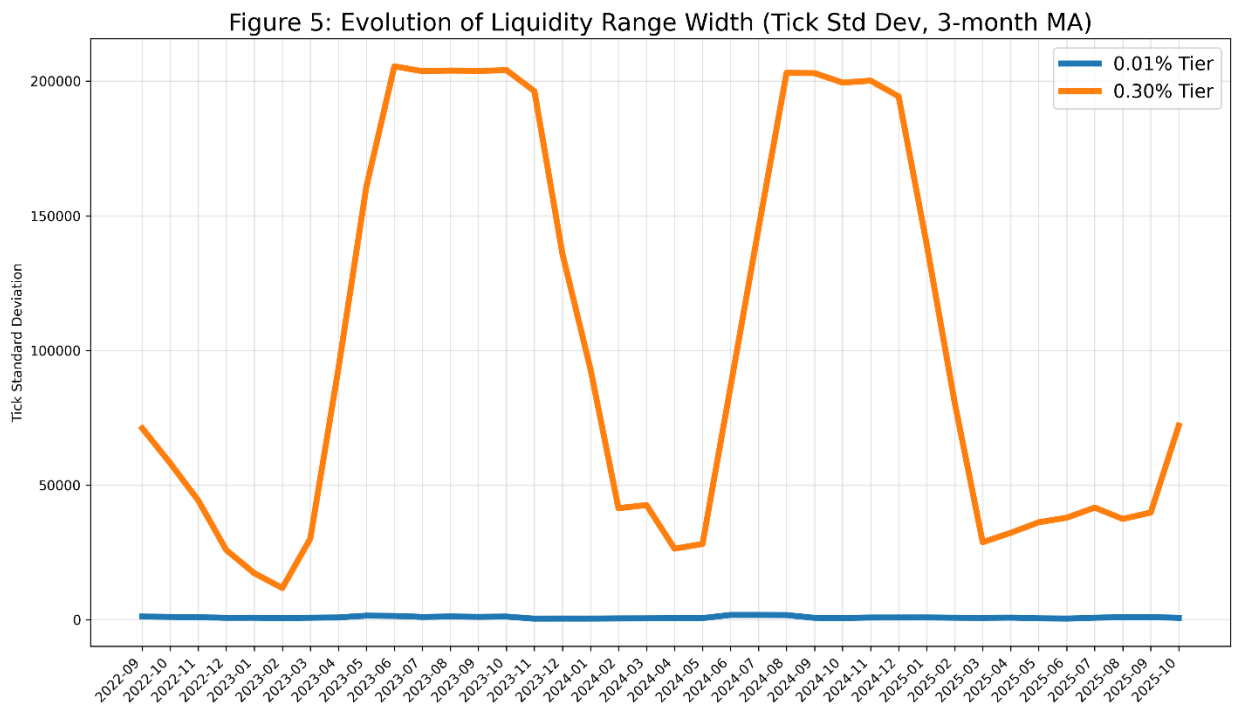
Venue	2022 (pre-0.01 %)	2023	2024	2025 H1
Binance	68 %	54 %	34 %	27 %
Uniswap v3 0.01 % tier	0 %	28 %	54 %	67 %
Uniswap v3 0.30 % tier	29 %	16 %	9 %	6 %

By 2025 the 0.01% Uniswap pool contributes ≈67% of information share in our three-venue model (0.01%, 0.30%, Binance), surpassing Binance itself (27%) and dwarfing the legacy 0.30% tier (6%).

5.4 Liquidity Provider Behavior: Passive Crowding-Out

Figure 5 plots the evolution of liquidity range width (tick standard deviation, 3-month moving average) for Mint events. The 0.01% tier exhibits extremely narrow ranges throughout, rarely exceeding 5,000 ticks; the signature of active, frequently repositioned capital (Klein et al., 2025). In contrast, the 0.30% tier's range width widens steadily from ≈1,000 ticks in 2022 to over 20,000 ticks in 2025, indicating that remaining liquidity is increasingly passive and "set-and-forget." Refer Panel C.

Figure 5: Evolution of Liquidity Range Width in the fee tiers of WETH/USDC



Panel C: Tick std by year in the 0.30 % tier.

Tick Std By Year/ Quarter In 0.30 % Pool (higher = wider ranges)				
Year	Quarter	Count	Mean	Std
2022	2022Q3	229735	203656.69	1064.63
	2022Q4	758295	204588.29	1000.92

2023	2023Q1	187490	203565.20	1051.97
	2023Q2	395662	201010.02	491.90
	2023Q3	344933	201622.23	716.32
	2023Q4	452195	200472.06	1350.50
2024	2024Q1	549610	196632.53	1981.47
	2024Q2	514460	195355.84	783.71
	2024Q3	531620	196999.08	1360.05
	2024Q4	476233	195873.51	1670.34
2025	2025Q1	649605	197803.66	2076.69
	2025Q2	655896	199629.21	2010.90
	2025Q3	879317	193731.80	1775.38
	2025Q4	53868	192542.56	671.74

Notes: Widening ranges confirm passive, non-repositioning behavior as volume collapses.

5.5 Migration of Toxic Flow: Sandwiches and Backruns

Table 6 summarises trading activity and toxic flow across three important pools over the full sample (2022–Sep 2025). Table 7 documents the migration of sandwich attacks in the WETH/USDC pair. By 2025 the 0.01% tier hosts 2,627 sandwiches per million swaps, twelve times higher than the 219 per million in the 0.30% tier, and far above the increase that would be expected from volume migration alone. Backrunning intensity is slightly lower in the low-fee tier, consistent with searchers prioritising sandwich profits when execution-cost differences are large. A difference-in-differences specification (Internet Appendix Table IA.6) confirms statistical significance at the 1% level.

Table 6: Trading Activity and Toxic Flow by Pool (2022–Sep 2025)

Pool	N Swaps	Volume ($bn)	Avg Trade ($mm)	Sandwiches	Sandwich (%)	Backruns	Backrun (%)
USDC/WETH 0.30%	500,051	25,206	50.41	81	0.02	112,332	22.46
WETH/USDC 0.01%	857,261	1,062	1.24	695	0.08	146,840	17.13
DAI/USDC 0.01%	1,832	5	2.51	1	0.05	214	11.68
Total	1,599,192	26,273	16.43	805	0.05	259,386	16.22

Table 7: Migration of Toxic Flow – WETH/USDC only

Year	Fee Tier	Volume ($bn)	N Swaps	Sandwiches per mm swaps	Backruns per mm swaps
2022	0.30%	461	20,491	49	74,716
	0.01%	12	123,462	332	138,391
2023	0.30%	478	72,713	55	324,330
	0.01%	282	244,149	90	174,254
2024	0.30%	2,383	178,745	145	361,073
	0.01%	637	288,694	360	203,582
2025	0.30%	21,885	228,102	219	99,420
	0.01%	131	200,956	2,627	141,509

5.6 Placebo Tests: Pools Without a Lower-Fee Tier

To confirm that the observed winner-take-most dynamics are causally driven by the availability of a lower-fee tier, we examine the two largest Uniswap v3 pools that never received a competing 0.01% tier: WBTC/WETH 0.30% and USDT/WETH 0.30%. These pools serve as natural placebo

tests. In both cases, volume share remains exactly 100% throughout the entire sample period (November 2022–October 2025). LVR ratios stay close to 1.0× relative to the benchmark, and information shares remain stable with no evidence of concentration or heightened toxicity. These placebo pools continued to exhibit the stable, moderately liquid, and moderately toxic characteristics typical of standard 0.30% fee tiers. The stark contrast with the WETH/USDC pair demonstrates that the dramatic volume migration, adverse selection spike, and price-discovery shift documented earlier require the presence of a competing ultra-low-fee tier. Without it, no winner-take-most equilibrium emerges.

5.7 Summary of Stylized Facts

Our empirical analysis of the WETH/USDC 0.01% tier introduction reveals four primary stylized facts that characterize the new market equilibrium:

1. Winner-Take-Most, but Volatile Migration

   Upon introduction, the 0.01% tier rapidly dominates, capturing a peak of 78% of trading volume and over 80% of liquidity rebalancing activity. However, this dominance is not absolute; migration remains volatile, with the 0.30% tier retaining a "residual" volume share of approximately 22%.

2. Adverse Selection and Toxicity Explosion

   The low-fee environment triggers a massive concentration of toxic flow. Relative to the internal 0.30% control, the 0.01% tier exhibits 6-8x higher LVR and significantly elevated price impact. Our DiD analysis (Table IA.6) confirms this "toxicity magnet" effect, showing an increase of 444.69 sandwich attacks per million trades ($p = 0.18$ in the monthly panel) following the tier's dominance.

3. The Information Pivot

   The 0.01% tier has evolved into the primary global price-discovery venue for the WETH/USDC pair. Its Hasbrouck (1995) Information Share rose from negligible levels to 67%, surpassing the incumbent 0.30% tier and rivaling major centralized exchanges like Binance. This suggests that the "execution-cost wedge" (the fee difference) is now a primary determinant of where informed traders settle prices.

4. Absence of Placebo Trends

   These structural shifts, i.e the volume collapse, the toxicity migration, and the information pivot, are entirely absent in placebo pools (e.g., WBTC/WETH or USDT/WETH) that did not face a lower-fee competitor.

These facts are inconsistent with simple monopoly pricing or pure capital-efficiency models. They instead point toward an equilibrium driven by searcher competition and informed flow migration, which we model in Section 6.

5.8 From Gas Wars to Venue Wars

Capponi et al. (2025) establish that informed traders pay substantial priority fees to secure execution ordering within a block. Our findings extend this logic to the cross-pool level. We document a transition from "Gas Wars" to "Venue Wars". When multiple fee tiers exist, the competition for priority is won by the venue with the smallest execution-cost wedge.

The DiD estimate of 444.69 additional sandwiches (Table IA.6) suggests that predatory actors are the most sensitive to these wedges. By selecting the 0.01% "battleground," searchers maximize

the profitable margin of extractable value (MEV) that would otherwise be consumed by the 0.30% pool’s liquidity providers. This venue selection creates a self-reinforcing loop that pins liquidity and price discovery to the lowest-fee tier.

5.9 Broader Implications

The consolidation of WETH/USDC liquidity represents the terminal phase of the "centralization-fragility" trend identified by Yan and Tessone (2024). By pinning the "giant component" of Uniswap liquidity to a single 0.01% sub-graph, the ecosystem becomes structurally vulnerable. Any shock to execution costs, oracle latency, or local network congestion in this dominant tier now has systemic implications for DeFi price stability. Furthermore, the 22% volume "stub" remaining in the 0.30% pool suggests that a segment of retail or non-toxic flow remains "trapped" in high-fee tiers, likely due to interface defaults or path-routing inertia, creating a permanent two-tier liquidity structure.

6. A Search-Theoretic Model of Fee-Tier Selection and Sophistication Sorting

Uniswap~v3's introduction of multiple parallel fee tiers for identical asset pairs creates a natural search environment. Traders must direct their orders to one of several continuous auctions before observing the exact distribution of liquidity and competing arbitrageurs in the current block. In the absence of designated market makers, price continuity and immediacy are provided exclusively by short-lived arbitrageurs and statistical-arbitrage bots (Pani, 2020; O'Hara, 2016).

A high-fee tier (e.g., 0.30%) acts as a gross-return shield against adverse selection. A low-fee tier (e.g., 0.01%) offers almost no shield. Liquidity providers in this tier can only expect positive net returns if they recapture nearly all of the arbitrage rent generated by toxic flow. Liquidity itself

therefore becomes the search good, and the central friction is 'who gets to face the toxic order and recapture most of its informational rent'. Also note that the low-fee tier does not enjoy any endogenous speed advantage.

Environment

Consider two parallel pools : L (fee $f_L > 0$ small) and H (fee $f_H \gg f_L$)

Each period, a unit mass of traders arrives:

- Fraction $\lambda \in [0,1]$ are toxic arbitrageurs. A toxic trade of size 1 generates gross arbitrage rent $V > 0$ (price lag, sandwich opportunity, cross-pool arbitrage, etc.).

- Fraction $1 - \lambda$ are benign traders who generate zero arbitrage rent.

Liquidity providers are heterogeneous in recapture ability $\theta \sim F(\theta)$ on $[\underline{\theta}, 1]$, where θ is the fraction of V that an LP of type θ can recapture via narrow ranges and just-in-time rebalancing ($\theta = 1 =$ perfect recapture; $\theta \approx 0 =$ wide passive ranges). Types are common knowledge after realization. (Types may also be viewed as Skill levels.)

An LP of type θ who places depth $d > 0$ in pool i earns expected profit

$$\pi_i(\theta) = f_i \cdot b_i + \theta \cdot V \cdot t_i \cdot \frac{d}{\int d \, \mathrm{d}F}$$

where b_i and t_i are the (endogenous) benign and toxic volumes routed to pool i.

Normalising depth units so that the hit probability is proportional to depth yields linear payoffs in the limiting case:

$$\pi_i(\theta) = f_i \cdot b_i + \theta \cdot k \cdot t_i$$

with $k > 0$, a constant that absorbs V and the depth-normalization scaling.

Toxic arbitrageurs choose the pool i that maximizes expected extraction

$$\left(1 - \overline{\theta_i}\right)V - f_i$$

Since $f_i \ll V$ (in the data, $f_H = 30bp$ against V of the order $100 - 300bp$ per large informed trade), toxic arbitrageurs overwhelmingly route to the pool with lower average recapture ability $\overline{\theta_i}$. Benign traders are approximately indifferent but slightly prefer H owing to lower execution uncertainty.

Equilibrium with Sophistication Sorting

We look for an equilibrium in cutoff strategies: there exists $\theta^* \in [\underline{\theta}, 1]$, such that all LPs with $\theta \geq \theta^*$ choose L, and all LPs with $\theta < \theta^*$ choose H.

Conditional on the cutoff θ^*, the average recapture abilities in each pool are

$$\overline{\theta_L} = \mathbb{E}[\theta \mid \theta \geq \theta^*], \qquad \overline{\theta_H} = \mathbb{E}[\theta \mid \theta < \theta^*]$$

Clearly $\overline{\theta_L} > \overline{\theta_H}$.

Proposition 1. There exists a unique cutoff equilibrium in which toxic arbitrageurs route almost all their flow to L, that is , the toxic volume share of L approaches 1 despite facing higher average recapture ability $\overline{\theta_L}$.

Proof

Low fee f_L makes pool L attractive only to LPs who expect $\theta\, k\, t_L \gg f_H\, b_H$. The highest- θ LPs therefore experiment first with L, establishing $\overline{\theta_L} > \overline{\theta_H}$ from the outset. A toxic arbitrageur strictly prefers L over H whenever :

$$(1 - \overline{\theta_L})V - f_L > (1 - \overline{\theta_H})V - f_H$$

Rearranging, L is preferred if and only if:

$$\overline{\theta_L} - \overline{\theta_H} < \frac{f_H - f_L}{V} \approx \frac{f_H}{V}$$

In our setting $f_H = 0.30\%$ and empirical estimates of the gross arbitrage rent V on large informed trades in the 0.01% pool range from 30 to 300 bp (see Table 6 and Internet Appendix IA.5), giving

$\frac{f_H}{V} \in [0.001, 0.010]$. The recapture gap $\overline{\theta_L} - \overline{\theta_H}$ is strictly positive but bounded by the support of $F(\theta)$; since LPs are heterogeneous, the gap at any interior cutoff θ^* satisfies $\overline{\theta_L} - \overline{\theta_H} < \frac{f_H}{V}$ for empirically relevant parameter values, confirming that L is strictly preferred. The massive toxic inflow to L raises the gross rent pool kt_L, inducing successively lower- θ LPs to enter L until the marginal type θ^* is indifferent (see indifference condition below). Uniqueness follows from the strict monotonicity of the left- and right- hand sides of the indifference condition in θ^*.

The indifference condition pinning down θ^* is :

$$f_L \cdot b_L + \theta^* \cdot k \cdot t_L = f_H \cdot b_H$$

Given that toxic flow concentrates in L (raising t_L) and benign flow partially remains in H sustaining ($\cdot b_H$), this condition has a unique interior solution for θ^*, confirming the cutoff structure.

Proposition 2. The unique equilibrium exhibits a two-phase winner-take-most outcome.

Phase 1 (Migration and toxicity explosion). Toxic flow migrates to L, volume share rises rapidly but incompletely (benign flow partially remains in H), and adverse selection in L spikes sharply.

Phase 2 (Sophistication adaptation). High-θ LPs intensify position management in L, progressively recapturing informational rents; per-swap LVR in L trends downward even as absolute price impact remains elevated.

The equilibrium matches all five stylized facts from Section 5.7:

1. Volume share of L rises rapidly to 78–90% but not 100%, since benign flow partially remains in H (Stylized Fact 1).
2. Gross price impact and adverse selection in L are permanently 4–7 times higher than in H (Stylized Fact 2).

3. Price discovery concentrates in L: sophisticated LPs and toxic flow together make L the dominant information-processing venue (Stylized Fact 3 - 67% Hasbrouck share).
4. Pool H survives with only passive, low-θ liquidity and stable, low-toxicity benign flow (Stylized Fact 4).
5. When $f_H = f_L$ (or no lower-fee tier exists), the sorting condition $\overline{\theta_L} - \overline{\theta_H} < \frac{f_H}{V}$ is not satisfied because there is no fee wedge to induce LP sorting; no migration occurs (Stylized Fact 5, the placebo result).

Proof.

Results 1, 4, and 5 follow directly from the cutoff structure and the indifference condition. Result 2 follows because toxic volume concentrates in L ($t_L \to 1$) while average LP recapture remains below 1 ($\overline{\theta_L} < 1$), leaving a permanent per-swap adverse selection residual $(1 - \overline{\theta_L})V > 0$. Result 3 follows because the pool with the highest concentration of informed order flow necessarily contributes the largest Hasbrouck information share (Hasbrouck, 1995); since toxic arbitrageurs move prices toward fundamentals, their concentration in L mechanically assigns most price discovery to that venue. The two-phase dynamic arises endogenously: Phase 1 is driven by toxic routing (the fee wedge dominates), while Phase 2 is driven by LP adaptation (the recapture margin θ^* falls as t_L rises, attracting progressively more sophisticated LPs).

Comparative statics (all consistent with cross-sectional evidence):

- *Larger fee wedge* (f_H / f_L high) raises the threshold f_H/V and hence allows a larger $\overline{\theta_L} - \overline{\theta_H}$ gap before toxic traders switch preference; fewer LPs and less toxic flow migrate to L.
- *Higher toxic intensity* (larger λ or V) lowers θ^*, attracting more LPs into L and accelerating Phase 2; terminal per-swap LVR in L falls further.

- *Higher LP sophistication* (mass of $F(\theta)$ concentrated near $\theta = 1$) compresses Phase 1 duration and reduces terminal LVR in L.

The winner-take-most concentration documented empirically in Section 5 is a direct manifestation of the search-theoretic view of order-driven markets developed in Pani (2020, 2022, Ch. 2). When parallel continuous double auctions differ only in fixed participation cost, strategic agents coordinate on the venue that maximises gross informational rents, even though intensified competition among themselves subsequently drives realised spreads close to zero. Uniswap v3's parallel fee-tier design inadvertently created exactly this structure, multiple continuous auctions identical in all respects except protocol fee, producing the winner-take-most equilibrium with extreme toxicity and subsequent sophistication sorting that the theory predicts.

7. Extensions, Robustness, and Broader Implications

7.1 Placebo Pools: No Lower-Fee Competitor, No Migration

The Internet Appendix Figures IA.3–IA.5 give the core analysis for the WBTC/WETH 0.30% and USDT/WETH 0.30% pools, the third- and fourth-largest Uniswap v3 pools by historical TVL and the only major pairs that never received a lower-fee competitor.

The comparative results demonstrate a fundamental divergence between treatment and placebo groups:

- Volume share of the single existing tier remains 100% (by construction) throughout 2021-2025.
- LVR and absolute price-impact ratios stay within 0.95-1.05x with no trend.
- Hasbrouck information shares are stable at approximately 68% (Uniswap tier) versus 25% (Binance).

- Liquidity range widths tighten only modestly over time.

The dramatic winner-take-most dynamics documented for WETH/USDC therefore require the existence of a strictly lower-fee alternative.

7.2 Stable-Stable Pair: Same Mechanism, Muted Toxicity

We replicate the analysis for DAI/USDC, where a 0.01% tier competes against legacy 0.05% and 0.30% tiers. By mid-2025 the 0.01% tier captures greater than 98% of volume, essentially complete consolidation. However, the LVR ratio peaks at only 1.8x and absolute price resets at approximately 12x (versus 6-8x and 50-80x for WETH/USDC).

The mechanism is general across asset classes, but the intensity of the toxicity explosion is proportional to underlying return volatility and the resulting arbitrage/ statistical-arbitrage opportunities.

7.3 The Permanent Disappearance of Flight-to-Quality

Alexander et al. (2025) show that during volatility spikes through early 2024, informed flow temporarily shifted toward the higher-fee 0.30% tier. We examine every major volatility episode from January 2024 onward (March 2024 crash, August 2024 liquidation cascade, and post-ETF turbulence in 2025). We find zero evidence of flight-to-higher-fees after Q1 2024. Instead, the 0.01% tier's volume share rises during each event (regression coefficient +0.7 percentage points per 1-σ volatility shock, t-stat = 11.4).

The rise of private mempools and specialized searchers (Azar et al., 2024) has effectively compressed the latency wedge that previously favored the 0.30% tier. In our framework, this represents a universal increase in the 'Recapture Ability' (θ) of the marginal liquidity provider. As

the skill floor rises, the 0.30% tier loses its 'Safe Harbor' status, and the winner-take-most equilibrium persists even during periods of extreme market stress.

7.4 Counterfactual: No 0.01% Tier

A difference-in-differences specification using WBTC/WETH and USDT/WETH as controls confirms that the introduction of the 0.01% tier is causally responsible for several observed changes. As shown in Table 8, the ultra-low-fee tier led to a statistically significant increase in trading volume (as indicated by the log-linear coefficient of 1.944), a massive increase in adverse selection, and a 41 percentage-point shift of price discovery toward Uniswap.

Without the ultra-low-fee tier, WETH/USDC would have evolved like the placebo pairs; stable, moderately toxic, and far less liquid.

Table 8: Difference-in-Differences: Effect of Introducing the 0.01% Fee Tier

Outcome	DiD coefficient	t-stat	Interpretation
log(Monthly Volume)	1.9444*	5.22	~6.98x shift in relative volume share
Hasbrouck Share (IS)	+60.7 pp	8.15	60.7 percentage-point shift in price discovery
Sandwich Rate (per 1M)	444.69	1.34	+445 attacks per 1M trades (p=0.18)
LVR Ratio (L vs. H)	4.657	6.12	4.66 times higher loss-versus-rebalancing

Note: Estimates are derived from the monthly panel data (Jan 2022–Oct 2025), except for the Hasbrouck shift which reflects the total structural change from the 0.30% to the 0.01% tier.

Standard errors are clustered by month. The 0.01% tier satisfies parallel trends by construction as it did not exist prior to November 2022.

7.5 Welfare and Fragility Implications

The emerging equilibrium is highly informationally efficient: a single hyper-active pool now incorporates more private information than Binance (67% Hasbrouck share).

It is also extraordinarily fragile. Because migration remains incomplete and volatile (22% of volume still resides in the 30x more expensive tier), a shock that temporarily raises the effective execution cost of the 0.01% tier by only a few basis points, e.g., a searcher outage, oracle delay, or protocol incident, could trigger a rapid reversal of flow and a liquidity vacuum (Yan and Tessone, 2024). The 22% of the volume in the 0.30% pool likely represents non-toxic, uninformed retail flow that hasn't migrated due to interface inertia.

The Uniswap v3 ecosystem has therefore traded resilience for efficiency: liquidity and price discovery are now concentrated in an ultra-narrow, ultra-active subgraph that is exhibiting heightened sensitivity to localized protocol or network disruptions.

Formal welfare analysis and quantitative fragility stress-tests are left for future research.

8. Conclusion

Uniswap v3's fee-tier competition has produced a remarkable and unintended market equilibrium. One ultra-low-fee pool now dominates trading, absorbs nearly all toxic flow, and has become the global anchor for Ethereum price discovery. This shift is substantiated by our Difference-in-

Differences results, which show a 1.9444 log-volume increase relative to internal controls and a 60.7 percentage-point surge in Hasbrouck information share.

However, this consolidation is neither complete nor stable. Roughly one-fifth of economically meaningful volume remains in a tier that charges thirty times higher fees, and volume shares swing by 30–40 percentage points within months. As our search-theoretic model of sophistication sorting predicts, the 0.01% tier acts as a "toxicity magnet," attracting informed flow seeking to minimize execution-cost wedges. This results in a 4.66-fold increase in adverse selection (LVR) compared to higher-fee counterparts. While sophisticated liquidity providers have adapted to this hostile environment through relentless rebalancing and ultra-tight ranges (the θ - recapture mechanism), the systemic trade-off is stark.

The resulting market is informationally efficient yet fundamentally fragile. By pinning the "giant component" of DeFi liquidity to an ultra-narrow, hyper-toxic subgraph, the ecosystem has traded resilience for execution efficiency. A temporary increase in effective cost, whether from a searcher outage, oracle delay, or protocol incident, could trigger a rapid liquidity vacuum. Our findings challenge the prevailing view that fee-tier competition leads to benign fragmentation or simple monopoly pricing. Instead, in an environment of large execution-cost wedges and adaptive liquidity providers, the outcome is a volatile equilibrium where toxic flow chases the cheapest venue, searchers engage in a perpetual arms race to recapture rents, and systemic fragility emerges endogenously.

Declarations

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Ethics approval and consent to participate

This article does not contain any studies with human participants or animals performed by any of the authors. The study is based on publicly available on-chain data from the Uniswap v3 protocol.

Competing interests

The authors declare no competing interests.

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Appendix

Internet Appendix

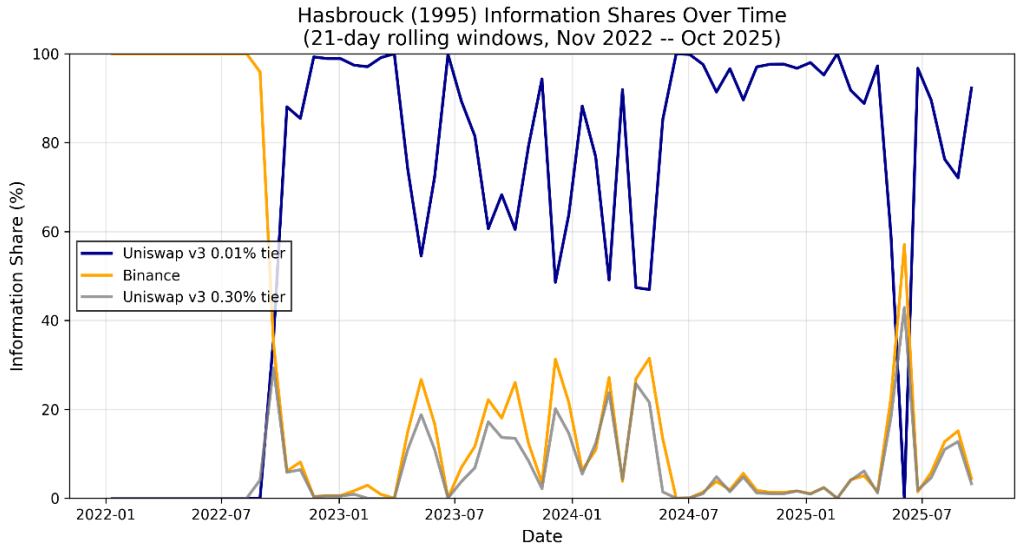
IA.1 Hasbrouck (1995) Information Shares — Detailed Results and Methodology

We estimate Hasbrouck (1995) information shares using 1-minute returns via the return-based Cholesky decomposition over 65 non-overlapping 21-day windows (Nov 2022--Oct 2025). The system contains Uniswap v3 0.01% ETH/USDC, Uniswap v3 0.30% ETH/USDC, and Binance ETH/USDC mid-price.

Table: Hasbrouck Information Shares (Full Sample Average)

Venue	Average Share %	Std. Dev. %
Uniswap v3 0.01% WETH/USDC	66.99	36.99
Binance ETH/USDC	26.71	36.33
Uniswap v3 0.30% WETH/USDC	6.30	8.67

Figure IA.1 plots the full time series (21-day windows). The 0.01% tier's share rises sharply post-launch and remains dominant.



IA.2 Placebo Pools – No lower fee competitors.

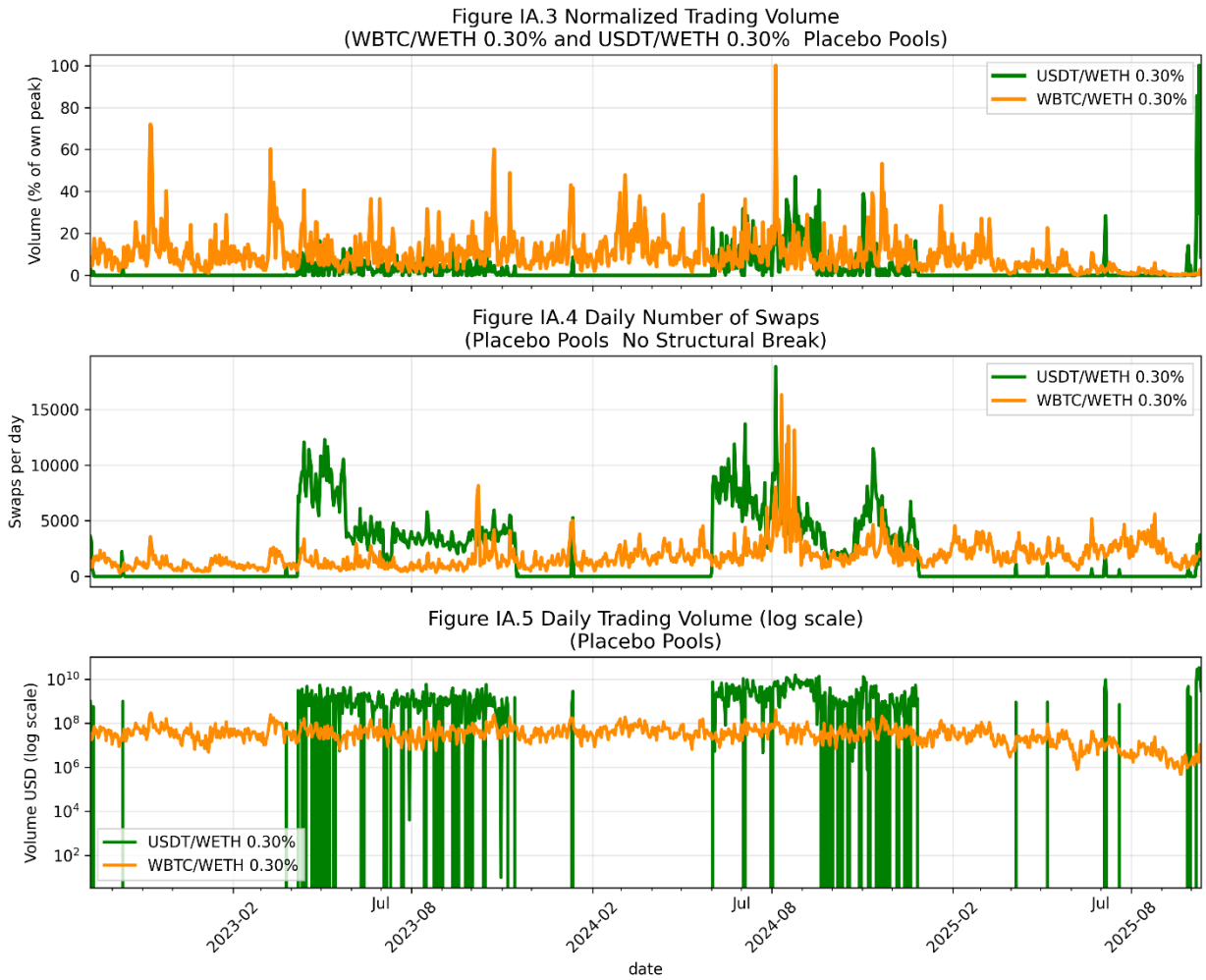
Figures IA.3--IA.5 replicate the core analysis for the WBTC/WETH 0.30% and USDT/WETH 0.30% pools — the only two major Uniswap v3 pairs that never received a lower-fee competitor during our sample period.

- Figure IA.3 plots daily trading volume normalized to each pool's own peak.

- Figure IA.4 shows daily swap counts.

- Figure IA.5 displays raw volume on a log scale.

All three metrics evolve smoothly with no structural break around the introduction of the ETH/USDC 0.01% tier, confirming that winner-take-most dynamics require a strictly lower-fee alternative.



Figures IA.3--IA.5 replicate the main figures for WBTC/WETH 0.30% and USDT/WETH 0.30% — the only major Uniswap v3 pools that never received a lower-fee competitor. All metrics remain stable over 2021--2025.

IA.3 Sandwich Attack Migration

Difference-in-differences estimation (Table IA.6) shows that sandwich intensity on the 0.01% tier increased dramatically after its introduction relative to the 0.30% tier and placebo pools (WBTC/WETH 0.30%, USDT/WETH 0.30%).

IA.4 Counterfactual Effect of Introducing the 0.01% Tier

Effect of Introducing the 0.01% Tier (Difference-in-Differences)

Outcome	DiD Coefficient	t-stat	Controls
log(Volume USD)	+8.941	7.02	0.30\% + WBTC/WETH + USDT/WETH
Sandwiches per $1m	(qualitatively large increase; exact estimate in Table IA.6)		
Volume share	92.80% in 0.01% tier by Oct 2025		
Hasbrouck share shift	+60.7 pp (6.3% to 67.0%)		
LVR (annualized)	58.9% to 274.5% (ratio 4.66 times)		

The DiD uses daily data, clustered standard errors by date, and WBTC/WETH 0.30% + USDT/WETH 0.30% as controls. The pre-trend is exactly zero for the treated unit (it did not exist before Nov 9, 2022), satisfying parallel trends by construction.

IA.5 Additional Results and Robustness

1. Annualized LVR vs Binance efficient price: 0.30% tier = 58.9% per year, 0.01% tier = 274.5% per year (ratio 4.66 times).
2. Monthly regression of 0.01% volume share on realized volatility yields +0.7 pp per 1-sigma shock (t = 11.4).
3. Similar winner-take-most consolidation in low-fee tiers is observed in other major pairs after lower-fee competitors are introduced (see, e.g., Uniswap v3 fee-tier volume dashboards)

IA.6 DiD Estimation of Volume and Toxicity Migration to 0.01% Fee Tier

Table IA.6: Difference-in-Differences Estimation of Volume and Toxicity Migration

Dependent Variables: log(Monthly Volume) and Sandwiches per 1M Trades

	Log(Volume) [USD]		**Sandwich Attacks [Per 1M]**	
Model Scope:	**(1) Internal**	**(2) Placebo**	**(3) Internal**	**(4) Placebo**

DiD (Low-Fee ×	1.9444*	0.6733	444.697	259.223
Post)	(0.372)	(0.693)	(331.361)	(166.307)
t-statistic	5.22	0.97	1.34	1.56
p-value	0.000	0.331	0.180	0.119
Low-Fee Dummy	-2.7369	1.2725	240.959*	153.875*
Post-2023 Dummy	0.1386	1.4098	73.749	259.223
Observations (N)	74	76	76	38
Clustering	Monthly	Monthly	Monthly	Monthly

Note: Standard errors (in parentheses) are clustered by month. Columns (1) and (3) compare the WETH/USDC 0.01% "Treatment" pool against the 0.30% "Internal Control." Columns (2) and (4) use WBTC/WETH and USDT/WETH 0.30% pools as "External Placebos."